Exercises I.1

Connor Mooneyhan

June 7, 2024

1.1. Locate a discussion of Russell's paradox, and understand it.

Proof. Done! □

1.2. Prove that if $\sim$ is an equivalence relation on a set S, then the corresponding family $\mathscr{P}_\sim$ defined in §1.5 is indeed a partition of S: that is, its elements are nonempty, disjoint, and their union is S.

Proof. If $S = \emptyset$, then $\mathscr{P}_\sim = \emptyset$ due to there being no elements with respect to which one can form equivalence classes, and $\emptyset$ does (vacuously) satisfy the conditions for a partition.

If S is nonempty, then given an equivalence class $[a]_\sim \in \mathscr{P}_\sim$, $a \in [a]_\sim$ since $a \sim a$.

For disjointness, let $b \in S$ such that $[b]_\sim \neq [a]_\sim$. We proceed to show that b is not related to a. Assume that $b \sim a$, and thus $a \sim b$. Then given $x \in [a]_\sim$, $x \sim a$, so by transitivity

$$x \sim a \sim b \implies x \sim b,$$

so $x \in [b]_\sim$. A similar argument follows for any $y \in [b]_\sim$, so we conclude that $[a]_\sim = [b]_\sim$ despite the assumption to the contrary. Thus b is not related to a.

Given $x \in [a]_\sim$, by definition $x \sim a$, so $a \sim x$ by symmetry. If $x \in [b]_\sim$, then $x \sim b$, so the transitivity of $\sim$ gives us

$$a \sim x \sim b \implies a \sim b,$$

contradicting our assumption. A similar argument can be made by swapping a and b, so we conclude that $[a]_\sim$ and $[b]_\sim$ are disjoint.

Finally, we show that $\bigcup \mathscr{P}_\sim = S$. Given $x \in \bigcup \mathscr{P}_\sim$, $x \in [a]_\sim$ for some $a \in S$, so by definition $x \in S$ since $[a]_\sim \subset S$. Assume, then, that $y \in S$. Then $y \in [y]_\sim \in \mathscr{P}_\sim$, we conclude that $y \in \bigcup \mathscr{P}_\sim$. □

1.3. Given a partition $\mathscr{P}$ on a set S, show how to define a relation $\sim$ on S such that $\mathscr{P}$ is the corresponding partition.

Proof. Given $x \in S$, let P_x be the unique $P_x \in \mathscr{P}$ such that $x \in P_x$. Define $\sim$ such that for any $a, b \in S$, $a \sim b \iff P_a = P_b$. We first show that this is an equivalence relation.

Given $a \in S$, clearly $P_a = P_a$, so $a \sim a$. Also, if $b \in S$ and $a \sim b$, then

$$P_a = P_b \implies P_b = P_a \implies b \sim a.$$

Finally, if $c \in S$ and $b \sim c$, then

$$P_a = P_b = P_c \implies P_a = P_c.$$

Now we show that $S/\!\sim\, = \mathscr{P}$. Let $[a]_\sim \in S/\!\sim$. We want to show that $[a]_\sim = P_a \in \mathscr{P}$, where P_a is defined as above. Given $x \in [a]_\sim$, $x \sim a$, so by definition of $\sim$, $P_x = P_a$. Thus since $x \in P_x$, $x \in P_a$. On the other hand, if $y \in P_a$, then since y is in only one member of $\mathscr{P}$ and $y \in P_y$, $P_y = P_a$ and thus $y \sim a \implies y \in [a]_\sim$.

Let $A \in \mathscr{P}$. Then $A = P_b$ for some b since A is nonempty. We want to show that $P_b = [b]_\sim \in S/\!\sim$. Given $x \in P_b$, since $x \in P_x$ and x is in only one member of $\mathscr{P}$, $P_x = P_b$. Thus $x \sim b$, so $x \in [b]_\sim$. If $y \in [b]_\sim$, then $P_y = P_b$, so since $y \in P_y$, $y \in P_b$. □

1.4. How many different equivalence relations may be defined on the set $\{1, 2, 3\}$?

Proof. Five. This can be found easily by simply determining the number of partitions. Here's the list:

$$\begin{aligned}
&\{\{1\}, \{2\}, \{3\}\} \\
&\{\{1\}, \{2, 3\}\} \\
&\{\{1, 3\}, \{2\}\} \\
&\{\{1, 2\}, \{3\}\} \\
&\{\{1, 2, 3\}\}
\end{aligned}$$

□

1.5. Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you attempt to use this relation to define a partition on the set? (Hint: Thinking about the second question will help you answer the first one.)

Proof. Let $S = \{1, 2, 3\}$, and define $\sim$ by

$$\begin{aligned}
1 &\sim 1 \\
1 &\sim 2 \\
2 &\sim 1 \\
2 &\sim 2 \\
2 &\sim 3 \\
3 &\sim 2 \\
3 &\sim 3.
\end{aligned}$$

It is easily checked that $\sim$ is reflexive and symmetric but not transitive. If you attempt to define a partition on S using $\sim$ in the usual way, then you get

$$\{[1]_\sim, [2]_\sim, [3]_\sim\} = \{\{1,2\}, \{1,2,3\}, \{2,3\}\},$$

which is not a partition since its members are not disjoint. $\square$

1.6. Define a relation $\sim$ on the set $\mathbb{R}$ of real numbers by setting $a \sim b \iff b - a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a 'compelling' description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by declaring $(a_1, a_2) \approx (b_1, b_2) \iff b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.

Proof. Given $a \in \mathbb{R}$, $a - a = 0 \in \mathbb{Z}$, so $a \sim a$. If $a \sim b$ for some $b \in \mathbb{R}$, then $b - a \in \mathbb{Z}$, so $a - b = -(b - a) \in \mathbb{Z}$ and thus $b \sim a$. Finally, if $c \in \mathbb{R}$ such that $b \sim c$, then $c - a = c - b + b - a \in \mathbb{Z}$ since both $c - b$ and $b - a$ are in $\mathbb{Z}$.

As for a compelling description, you could characterize each member of $\mathbb{R}/\sim$ as simply $\mathbb{Z}$ translated by some $x \in [0, 1)$.

Now we shift our focus to $\approx$, where we aim to show not only that $\approx$ is an equivalence relation on its own, but that it *inherits* the properties of reflexivity, symmetry, and transitivity from $\sim$. To do this, we notice that we can recharacterize the definition of $\approx$ as

$$(a, b) \approx (c, d) \iff a \sim c \text{ and } b \sim d.$$

This is clear from the definitions of $\approx$ and $\sim$.

First, we notice that reflexivity follows from the fact that given $(a, b) \in \mathbb{R}^2$, $a \sim a$ and $b \sim b$, so $(a, b) \approx (a, b)$. Then assume that $(a, b) \approx (c, d)$ for some $(c, d) \in \mathbb{R}^2$. Then $a \sim c$ and $b \sim d$ by our re-definition, so $c \sim a$ and $d \sim b$, implying that $(c, d) \approx (a, b)$. Finally, given $(e, f) \in \mathbb{R}^2$ such that $(c, d) \approx (e, f)$, we have

$$a \sim c \sim e \implies a \sim e$$

and

$$b \sim d \sim f \implies b \sim f,$$

so $(a, b) \approx (e, f)$.

We carry out the proof in this way so that it is apparent that nothing in the proof depends on the particulars of $\sim$. This hints at the appropriateness of characterizing $\approx$ as an instance of a "product relation", namely

$$\approx = \sim \times \sim .$$

Our proof is sufficient to show that any such product of an equivalence relation with itself is itself an equivalence relation.

Enough generalization. A compelling description for $S/\approx$ is similar to that of $S/\sim$, where for each member of $S/\approx$, we imagine the lattice $\mathbb{Z}^2$ translated by $(x, y) \in [0, 1) \times [0, 1)$. $\square$